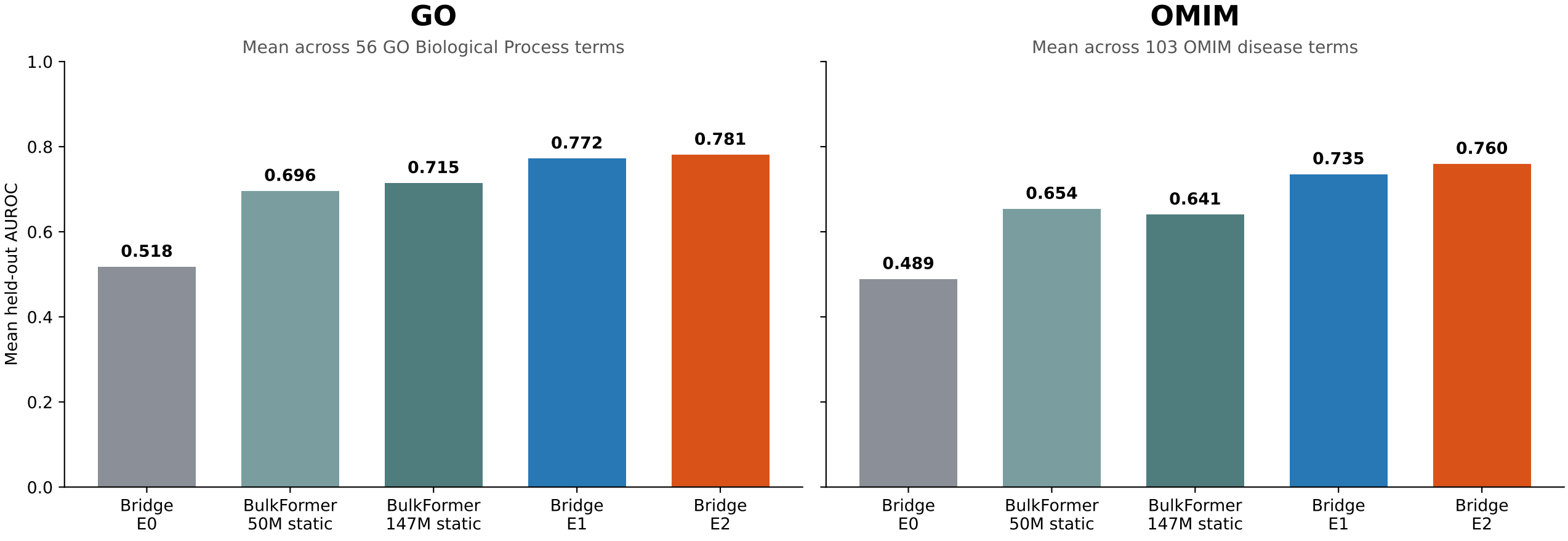Single-gene biology from frozen gene representations